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ORIGINAL PAPER

TSNet with sigma-adaptive spectral sampling for tea moisture monitoring

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Abstract

Accurate, non-destructive estimation of leaf moisture during tea withering is foundational for closed-loop control and quality assurance in industrial tea processing. Although near-infrared spectroscopy is widely adopted for this purpose, field deployment remains fragile: leaf folding/occlusion and pose drift introduce spectral instability and necessitate frequent recalibration, undermining robustness and scalability. To overcome these limitations, we develop SpecTea, an end-to-end system that integrates robust spectral acquisition, sequence modeling, and real-time inference. The method exploits the dynamics of withering to enhance spectral separability and reduce information entropy, and introduces a SASS strategy that combines spatially adaptive five-point segmentation with dynamic Poisson-disk sampling to select informative regions. A custom spectroscopic apparatus secures representative measurements, which are modeled by TSNet, a sequence architecture that fuses self-attention with GRUs for moisture prediction. On held-out tests, TSNet achieves a correlation coefficient of 0.9917, an RMSE of 2.7612, and an RPD of 2.7552, surpassing traditional machine-learning baselines and recent deep models. The SASS-enabled acquisition consistently reduces inter-spectral variance, improving generalization across batches and processing conditions. When deployed on the withering line, the system attains an average error of 3.89%, comfortably meeting the $\leq 5\%$ industrial threshold and demonstrating stable online performance despite leaf pose and occlusion variations. The approach also lowers recalibration frequency, reducing operational burden and downtime. By unifying adaptive sampling with attention-based sequence learning in a deployable pipeline, SpecTea delivers a practical, scalable solution for real-time NIR moisture sensing, offering a transferable blueprint for robust quality control in tea manufacturing and related agri-food processes.

Keywords Near-infrared spectroscopy · Tea withering · Moisture detection · Sequence model · Attention mechanism

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Introduction

Tea is one of China's most important economic crops and a globally traded beverage commodity; its quality directly shapes market competitiveness and consumer demand. As shown in Fig. 1, which outlines the key stages of tea production, among all physicochemical attributes, moisture content is a critical control variable throughout the withering–rolling–drying pipeline, influencing enzymatic activity, aroma development, color retention, and texture [1]. Overmoist leaves tend to yellow, deteriorate, and lose desired fragrance notes, whereas under-moist leaves are prone to scorching and shrinkage during thermal processing, ultimately degrading mouthfeel and cup quality [2]. Consequently, reliable, real-time regulation of moisture is central to quality assurance and food safety in modern tea manufacturing. In practice, however, tea processing is a complex, dynamic environment. Conveyor motion, leaf folding and occlusion, pose drift, and batch-to-batch heterogeneity introduce uncertainty that manifests as spectral noise, variability, and weak or non-identifiable features in field-acquired signals [3]. Moreover, building robust predictive models requires (i) selecting appropriate regions of interest on irregular, curved leaf surfaces and (ii) ensuring consistent reflectance measurements across time and batches—both of which remain challenging on industrial lines. These technical hurdles compound with a pragmatic constraint: commercial moisture analyzers frequently exceed 8% error at high moisture levels, and once the deviation surpasses 6.5%, the resulting quality loss is unacceptable for production use [4]. Together, these limitations underscore the need for a moisture detection system that is not only accurate but also stable and deployable under realistic operating conditions.

Recent advances in in situ instrumentation (e.g., the technique reported by Nipan Das et al. [5]) offer promising avenues for real-time moisture sensing, but adoption on factory floors is still hampered by system complexity and integration overheads. By contrast, near-infrared spectroscopy (NIRS) provides a compact, non-destructive, and scalable solution that aligns well with on-line monitoring requirements [6]. Yet, NIRS in tea production faces persistent barriers: the shape, angle, and surface microstructure of fresh leaves induce strong spatial variability in reflectance, so spectra from different sampling regions—even within the same batch—can diverge markedly, degrading model accuracy and hindering large-scale deployment [7]. Further, chemical composition and tissue structure vary across cultivars and seasons, and moisture distributions shift with processing stages; coupled with ambient temperature and humidity fluctuations, these factors inject noise and drift into the spectra, destabilizing predictions in the wild [8–10]. Motivated by these challenges, our objective is to develop a field-ready tea-moisture sensing pipeline that is accurate, stable, and deployable under real production constraints. Specifically, we couple label-free sigma-adaptive sampling (SASS), a compact sequence model (TSNet), and end-to-end on-line integration (SpecTea) for industrial validation.

To address these issues, this study proposes a field-ready pipeline that couples adaptive sampling with sequence

Fig. 1 Tea Production Process



(a) Tea Picking (b) Leaf Drying

(c) Tea Processing (d) Tea Brewing

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modeling for robust moisture prediction from NIR spectra. First, we introduce a Sigma-Adaptive Spectral Sampling (SASS) strategy that partitions a leaf surface using spatially adaptive five-point segmentation and augments it with dynamic Poisson–disk sampling. This hybrid design concentrates measurements in regions exhibiting higher local variability while reducing redundancy in homogeneous areas, thereby improving representativeness and mitigating artifacts from curvature and occlusion. Second, we develop TSNet, a compact sequence model that combines self-attention with gated recurrent units (GRUs) to learn temporally coherent, noise-robust spectral representations. To further suppress environmental interference, we incorporate denoising and wavelet-based feature enhancement and employ a lightweight multi-scale CNN module to bolster resilience to distributional shifts.

Finally, we integrate the above components into a real-time, online moisture detection system suitable for withering-line deployment. Extensive experiments show that TSNet achieves a correlation coefficient of 0.9917, an RMSE of 2.7612, and an RPD of 2.7552 on held-out tests, consistently outperforming traditional machine learning and recent deep models. When deployed on-line, the system attains an average detection error of 3.89%, comfortably below the 5% threshold used in fresh tea quality assessment, demonstrating practical feasibility for industrial settings. The main contributions are summarized as follows:

1) Sigma-Adaptive Spectral Sampling : A hybrid sampling strategy that integrates spatially adaptive five-point segmentation with dynamic Poisson–disk sampling to allocate measurements where they matter most.

2) TSNet for sequence modeling: An attention–GRU architecture tailored for NIR spectra that delivers state-of-the-art accuracy (corr. 0.9917, RMSE 2.7612, RPD 2.7552) on the proposed dataset.

3) Real-time deployment: An engineered, online moisture detection system that sustains 3.89% average error on the withering line, meeting industrial acceptance criteria and enabling reliable process control.

The abbreviations used throughout this paper are summarized in Table 1. Section "Introduction" introduces the background and recent progress of teamoisture detection. Section "Related work" reviews related moisture-estimation algorithms and sampling strategies for spectral sensing. Section "System design" presents the overall design of the proposed SpecTea system, including the hardware configuration and the Sigma-Adaptive Spectral Sampling (SASS) workflow. Section "Experiment" reports the experimental evaluation of the proposed TSNet model, including comparative results against baseline methods, crossvalidation, and ablation studies. Finally, Section "Conclusion" concludes the paper and discusses limitations as well as future research directions. Related work.

Table 1 List of abbreviations used in this paper

Abbreviation	Definition
NIR	Near-infrared
MAE	Mean absolute error
RMSE	Root mean squared error
CI	Confidence interval
S-G	Savitzky–Golay smoothing (Savitzky–Golay filter)
WT	Wavelet Transform
MSC	Multiplicative Scatter Correction
SNV	Standard Normal Variate
1st Derivative	First derivative (spectral derivative preprocessing)
2nd Derivative	Second derivative (spectral derivative preprocessing)
Vector	Vector normalization (L2 normalization)
Max–Min	Min–max normalization (max–min scaling)
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
MLP	Multi-layer perceptron
PLSR	Partial least squares regression
LS-SVR	Least-squares support vector regression
RPD	Ratio of performance to deviation
TSNet	Tea Spectrum Network (proposed model)
SASS	Sigma-Adaptive Spectral Sampling (proposed strategy)

Related work

Moisture detection in tea withering

Tea withering is a pivotal stage in tea processing in which controlled moisture reduction triggers coupled physical and biochemical transformations within leaf tissues. Moisture content is therefore a primary process indicator for evaluating withering progress [11]. Accurate, real-time moisture sensing enables dynamic set-point adjustment, stabilizes product quality, and enhances flavor consistency [12]. Beyond quality assurance, reliable online monitoring improves line efficiency and underpins the digitalization and intelligent automation of modern tea manufacturing.

Despite meaningful advances, current approaches face intertwined limitations in accuracy, operational flexibility, and cost. The spectral method of Nipan Das *et al.* achieves 85.41% classification accuracy with SVMs—below the $\geq 95\%$ reliability typically required for industrial automation

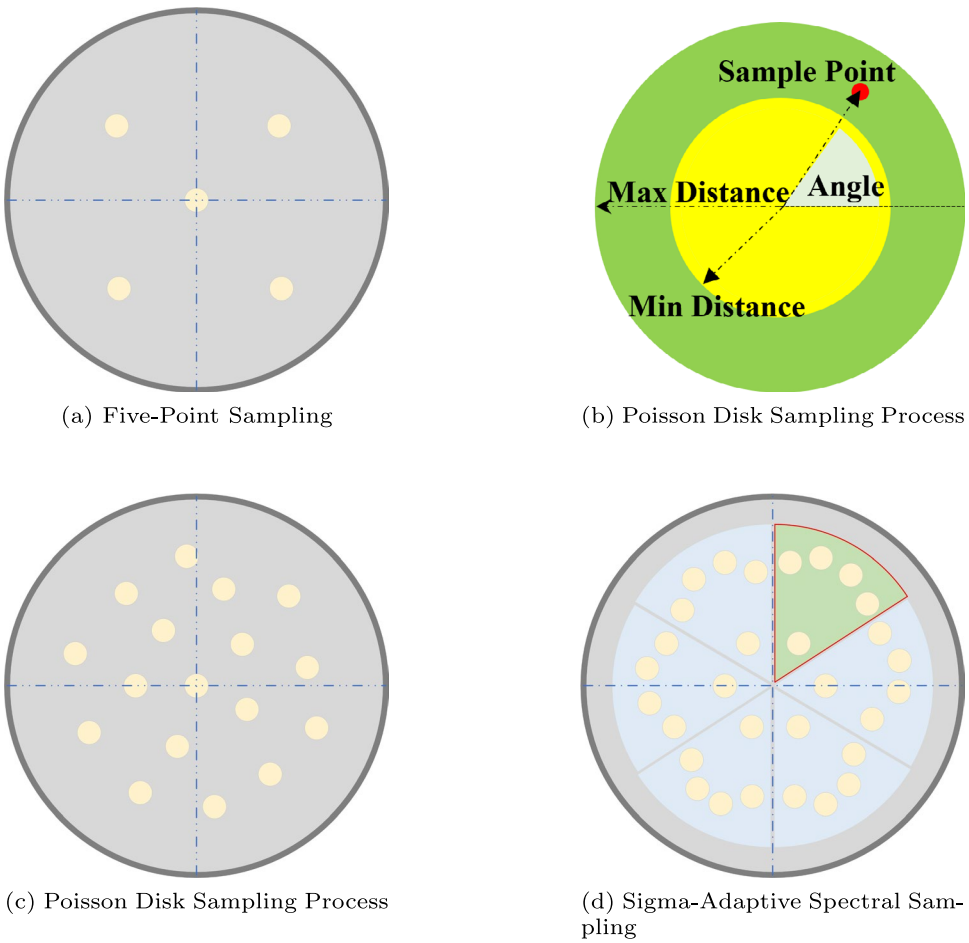
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[5]. Shyh Yau Tein *et al.* report excellent precision using 10.3 GHz microwave reflection (maximum error 0.52%), but the rigid, fixed-container protocol conflicts with moving-belt production [13]. J. A. Ling *et al.* obtain 99.59% volumetric accuracy with a coaxial monopole sensor, yet dependence on high-cost vector network analyzers impedes scale-out [14]. To address these gaps, we propose an edge-deployable, hybrid sensing framework that integrates spectral feature fusion and adaptive environmental compensation, high-precision network architecture, and cost-efficient hardware co-design. In deployment, the system attains a 3.89% moisture error—comfortably within the 5% industrial tolerance—while enabling unconstrained, on-line operation and achieving a 97.6% cost reduction relative to [14]. This corresponds to a 29.8% relative error reduction compared with the 14.59% error rate implied by [5], removes the spatial constraints inherent to [13], and resolves the practical precision–cost trade-off that has limited large-scale adoption.

Sampling method

Sampling strategy directly determines the accuracy and efficiency of data acquisition in tea-moisture detection, because it governs three core properties of a dataset: spatial coverage, adaptability to local heterogeneity, and statistical representativeness. Classical and modern schemes emphasize these properties to different degrees. The five-point sampling method, widely used in biological and agricultural studies [15], places one point at the region center and four along the diagonals at equal distances. As illustrated in Fig. 2(a), this geometry yields balanced and symmetric coverage with minimal operational cost, making it suitable for regular, approximately homogeneous areas in which moisture gradients are weak. However, its fixed layout does not adapt to local variability: when moisture exhibits anisotropy, localized hotspots, or edge effects, the central-plus-diagonal pattern undersamples informative subregions and oversamples redundant ones. Consequently, the method can bias reflectance estimates and degrade model accuracy, motivating the need for adaptive or density-aware sampling strategies in realistic, spatially heterogeneous settings.

Poisson–disk sampling is a stochastic scheme that generates evenly spaced, non-overlapping points [16–18]. As illustrated in Fig. 2(b)–(c), enforcing a minimum inter-sample distance yields near-uniform coverage while retaining randomness,



(a) Five-Point Sampling

(b) Poisson Disk Sampling Process

(c) Poisson Disk Sampling Process

(d) Sigma-Adaptive Spectral Sampling

Fig. 2 Specifications of Experimental Sampling Area

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which reduces clustering artifacts and improves statistical efficiency. Recent variants leveraging blue-noise optimization and GPU acceleration further increase sampling throughput and spatial realism [19]. However, the standard formulation uses a fixed exclusion radius and thus does not adapt sampling density to spatial heterogeneity, an important requirement in tea-moisture detection, where local variance and anisotropic gradients carry most of the predictive signal. Consequently, uniform-spacing priors can oversample homogeneous zones and undersample informative subregions, motivating variable-density or feature-aware extensions in practical deployments.

To address the limitations of fixed-layout and uniform-density schemes, we introduce SASS, which aligns sampling density with local spectral variability. The core idea is to partition the sampling field into angular sectors and maintain, within each sector, an exponential-moving estimate of moisture variance. SASS then adapts the local sampling radius according to this variance, allocating denser samples where variability is high and sparser samples where the field is homogeneous. Within every sector, a five-point pattern is instantiated using the sector-specific radius, marrying the geometric regularity of structured layouts with the responsiveness of data-driven adaptivity. As depicted in Fig. 2(d), this design improves representativeness and reduces redundancy by concentrating measurements on informative subregions, thereby enhancing sampling efficiency and the reliability of tea-moisture detection.

System design

A tea-moisture detection system, termed SpecTea, was developed to operationalize the SASS strategy. As shown in Fig. 3, the pipeline comprises three modules: Profiling, Training, and Deployment. Profiling: a custom spectral acquisition device implementing the SASS protocol performs adaptive sampling of reflectance signals with enhanced information density; the resulting spectra are standardized and partitioned to construct a high-quality dataset for model learning. Training: a predictive architecture, TSNet, integrates self-attention with gated recurrent units (GRUs) to capture global dependencies and temporal dynamics in NIR sequences; comparative experiments against conventional machine-learning approaches demonstrate superior prediction accuracy. Deployment: the trained TSNet is encapsulated as an application programming interface (API) and integrated into a real-time data pipeline that ingests fiber-optic spectra and executes on-device inference; a human–machine interface (HMI) visualizes predicted moisture values, and a control system enables real-time feedback to external equipment. These components deliver a field-ready solution that links adaptive sampling with robust sequence modeling for reliable online moisture estimation.

Profiling design

Spectral profiling environment

The tea spectral sampling environment in this study was manually constructed to ensure controlled yet realistic conditions. A Y-type optical fiber was employed as the transmission medium for near-infrared light, as its beam-splitting and integration capability allows stable propagation through the incident, reflection, and receiving ends. The fiber probe was positioned vertically above the tea leaves, such that the incident light was projected perpendicularly onto the tea surface, and the reflected light was collected along the original optical path.

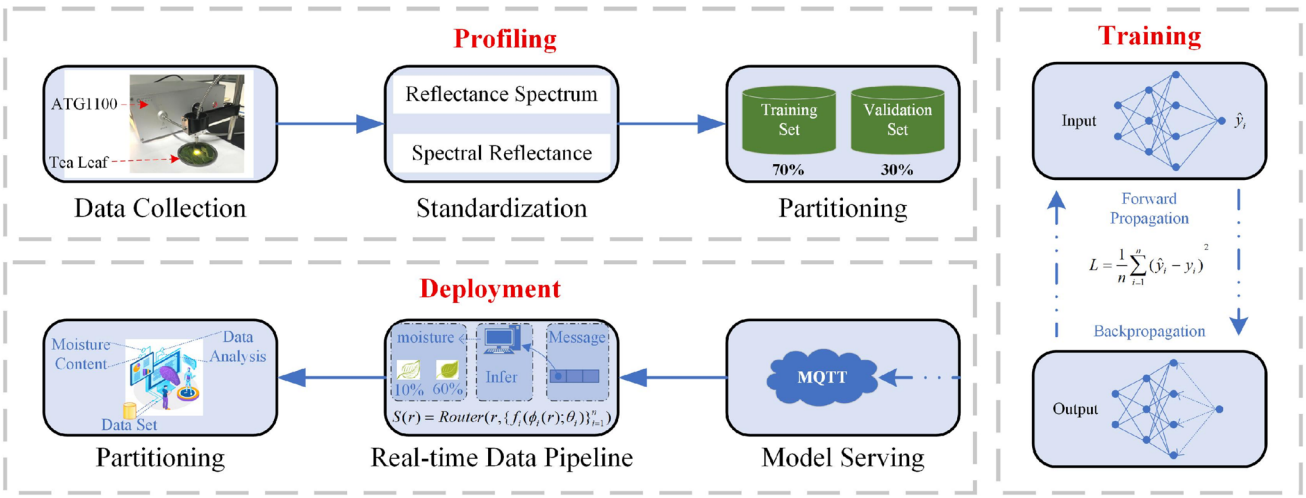


Fig. 3 Overview of the SpecTea

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The light source power was set to 25 W, and the tea tray used had a diameter of 9.5 cm. Each tray was fully loaded with tea leaves, with the total weight adjusted according to moisture content (2–4 g). The optical fiber was fixed onto a stand to maintain a constant probe-to-sample distance of 4.2 cm, which produced a light spot of approximately 2.6 cm in diameter. To better simulate real-world uncertainties, the experiment was conducted under natural indoor illumination rather than in a dark-room, while maintaining a stable ambient temperature of 24 °C. This setup ensured both reproducibility and relevance to practical tea-processing conditions.

Sigma-adaptive sampling strategy

Moisture content often exhibits spatial heterogeneity across the tea tray, requiring an adaptive sampling mechanism to improve measurement efficiency. Importantly, in real industrial settings, ground-truth moisture cannot be measured at every location; therefore, the proposed sigma-adaptive sampling is label-free at deployment and is driven only by spectral variability (or prediction uncertainty) computed from non-destructive NIR spectra. Ground-truth moisture is used only offline for model training/calibration and is not required during on-line sampling.

We model the tray as a circular domain $\mathcal{D}$ of radius R and partition it into M equal angular sectors:

$$\Omega_i = \{(r,\theta) \mid 0 \le r \le R,\ \tfrac{2\pi(i-1)}{M} \le \theta < \tfrac{2\pi i}{M}\}, \quad i = 1,\dots,M. \tag{1}$$

This partition enables localized analysis of variability. For each sector Ω_i, we estimate the red spectral variance using an exponential moving update:

$$\mathscr{V}_i^{(t)} = \alpha\, \mathscr{V}_i^{(t-1)} + (1-\alpha)\, \widehat{\mathscr{V}}_i^{(t)}, \quad \alpha \in [0,1), \tag{2}$$

where $\widehat{\mathscr{V}}_i^{(t)}$ is computed from the acquired spectra in sector Ω_i at iteration t (no moisture labels involved). Let $\mathrm{s}_{i,k}^{(t)} \in \mathbb{R}^L$ denote the preprocessed NIR spectrum at the k-th sampled point in Ω_i. We compute the sector-wise spectral variability on a feature mapping $\phi(\cdot)$ as

$$\widehat{\mathscr{V}}_i^{(t)} = \frac{1}{K}\sum_{k=1}^{K} \left\| \phi\left(\mathrm{s}_{i,k}^{(t)}\right) - \frac{1}{K}\sum_{k'=1}^{K} \phi\left(\mathrm{s}_{i,k'}^{(t)}\right) \right\|_2^2, \tag{3}$$

which provides a stable, label-free indicator of local heterogeneity.

To adjust sampling density accordingly, we define the adaptive radius in sector Ω_i as

$$r_i^{(t)} = r_0\left(1 + \kappa\, \frac{\mathscr{V}_i^{(t)}}{\max_j \mathscr{V}_j^{(t)} + \delta}\right), \tag{4}$$

where δ is a small constant for numerical stability. This ensures denser sampling in high-variance regions and avoids redundancy in uniform areas. To prevent spatial clustering, we enforce a Poisson-disk (minimum-distance) constraint when inserting candidate points into the global sampling set $\mathcal{S}$. A candidate point p is accepted only if

$$\min_{\mathrm{q}\in\mathcal{S}} \|\mathrm{p} - \mathrm{q}\|_2 \ge d_i^{(t)}, \tag{5}$$

where we simply set the sector-adaptive minimum separation as $d_i^{(t)} = r_i^{(t)}$ (same scale as the sigma-adaptive radius). If the constraint is violated, p is jittered/resampled within Ω_i until Eq. 5 holds (with a small max-try limit).

Within each sector, we adopt a five-point scheme to balance coverage and efficiency. The centroid $\mathrm{p}_{i,0}$ is first placed, followed by four additional points distributed along angular increments:

$$\theta_{i,k} = \frac{2\pi(i-1)}{M} + \frac{k}{4}\left(\frac{2\pi}{M}\right), \quad k = 1,2,3,4, \tag{6}$$

$$\mathrm{p}_{i,k} = \mathrm{p}_{i,0} + r_i^{(t)} \begin{bmatrix} \cos\theta_{i,k} \\ \sin\theta_{i,k} \end{bmatrix}, \quad k = 1,\dots,4. \tag{7}$$

After acquiring spectra at these points, the sector variance $\widehat{\mathscr{V}}_i^{(t)}$ is updated using Eq. 3, and the radius $r_i^{(t+1)}$ is recomputed via Eq. 4. The process iterates until the maximum radius change across all sectors falls below a threshold ε. The complete procedure is summarized in Algorithm 1.

Algorithm 1 Sigma-Adaptive Spectral Sampling.

Require: Base radius r_0, gain κ, sectors M, variance decay α, convergence ε
Ensure: Sampling set $\mathcal{S}$
Initialize $\mathscr{V}_i^{(0)}$ via a coarse spectral pre-scan
$t \leftarrow 0$, $\Delta \leftarrow \infty$, $\mathcal{S} \leftarrow \emptyset$
while $\Delta > \varepsilon$ **do**
  for $i = 1 \to M$ **do**
    Compute $r_i^{(t)}$ by Eq. 4
    Generate five points $\{\mathbf{p}_{i,k}\}_{k=0}^{4}$ by Eq. 7
    Add $\{\mathbf{p}_{i,k}\}$ to $\mathcal{S}$
    Acquire NIR spectra $\{\mathbf{s}_{i,k}^{(t)}\}_{k=0}^{4}$ at $\mathbf{p}_{i,k}$ (non-destructive)
    Compute $\widehat{\mathscr{V}}_i^{(t)}$ from $\{\mathbf{s}_{i,k}^{(t)}\}$ using Eq. 3
    Update $\mathscr{V}_i^{(t+1)} \leftarrow \alpha\mathscr{V}_i^{(t)} + (1-\alpha)\widehat{\mathscr{V}}_i^{(t)}$
  end for
  $\Delta \leftarrow \max_i |r_i^{(t+1)} - r_i^{(t)}|$
  $t \leftarrow t+1$
end while
return $\mathcal{S}$

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Dataset

The tea–spectrum acquisition campaign was conducted in a temperature-controlled laboratory from October to November 2023 using 2,000 g of fresh, unprocessed tea leaves. The workflow produced (i) one dark-reference dataset, (ii) eight standard reference spectra acquired at distinct integration times (ms), and (iii) 215 tea-sample batches, each yielding seven spectral curves (i.e., 215×7 spectra). This design supplies stable references and repeated measurements per batch for calibration and validation. To span a broad moisture range, samples were first air-withered indoors under natural conditions; a subset was then deliberately dried below the minimum withering threshold (i.e., $< 40\%$) to ensure inter-batch comparability and to enrich low-moisture coverage for training. Consequently, the overall moisture content across samples ranges from 0% to 80%, providing a dataset suitable for learning robust moisture-prediction models.

Because sampling height and spectrometer integration time affect raw intensity readings, we do not use raw spectra directly. Instead, measurements are converted to spectral reflectance via Eq. 7 before modeling:

$$\rho = \frac{X_c - X_d}{X_s - X_d} \times 100\%, \tag{8}$$

where X_c is the current sample spectrum, X_s the standard spectrum, and X_d the dark current. The spectrometer (ATP8600) provides 256 effective channels; thus X_c, X_s, X_d, and ρ are vectors in $\mathbb{R}^{1\times 256}$. Elements of X_c, X_s, and X_d are intensity values (instrument units), while ρ is expressed in percent.

As summarized in Fig. 4, each of the 215 samples has a moisture label t paired with seven original spectra x. After transformation by Eq. 8, the reflectance tensor per sample has size 7×256. To ensure consistency, each sample is matched to the standard spectrum corresponding to the spectrometer's integration-time setting k used during acquisition, thereby improving comparability and overall dataset reliability.

Training design

This study proposes Tea Spectrum Network (TSNet), a specialized moisture prediction model tailored for tea spectrum datasets, based on sequence modeling techniques and the performance of the Transformer model in spectral analysis. The proposed model integrates the self-attention mechanism for effective feature extraction and the RNN for sequential data modeling, demonstrating outstanding inference performance in tea spectrum sequence analysis tasks. The model's core procedure is formalized in Algorithm 2, which outlines the iterative training process encompassing

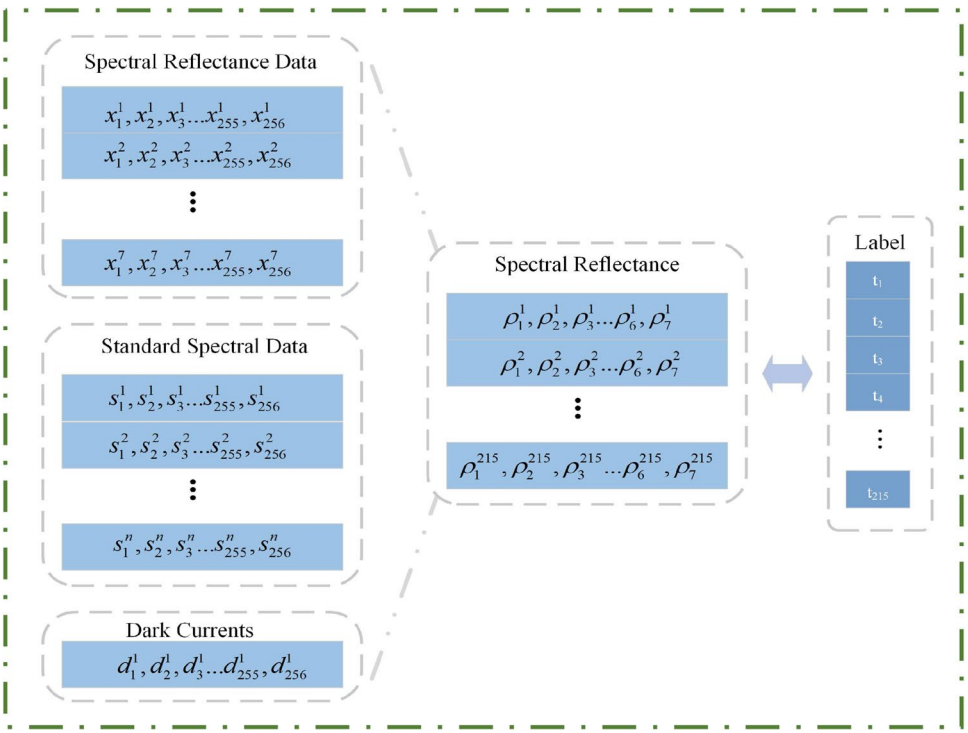


Fig. 4 Construction pipeline of the tea spectral database

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spectral normalization, dual-branch feature fusion (GRU-based temporal modeling and spatial attention), and residual-aided prediction.

Algorithm 2 Training Process of TSNet.

Require: Tea moisture sequences $\{X_t\}_{t=1}^{T}$
Ensure: Predicted sequences $\widehat{Y}$
1: Initialize network parameters θ.
2: **for** each epoch **do**
3: **Preprocessing:** Normalize the NIR signals:

$$X'_t = \mathcal{P}(X_t).$$

4: **Feature Extraction:** Compute self-attention components:

$$Q, K, V = W_Q X'_t, W_K X'_t, W_V X'_t.$$

5: **Dual-Branch Fusion:** Extract temporal and spatial features:

$$F_{\text{fused}} = Norm(\text{GRU}(X'_t) + \text{Mean}(X'_t)).$$

6: **Prediction:** Generate output via residual modeling:

$$\widehat{Y} = \text{TCResConv}(F_{\text{fused}}).$$

7: **Optimization:** Compute loss and update parameters:

$$\mathcal{L} = \mathcal{L}(\widehat{Y}, Y), \quad \theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}.$$

8: **end for**

Given the remarkable capability of the self-attention mechanism in Transformer models for feature extraction in sequential data, the TSNet model integrates this mechanism to enhance the precision of spectral feature band extraction for tea samples. By leveraging self-attention, the model effectively focuses on key spectral features, improving the accuracy of moisture prediction.To further process and integrate these extracted features, TSNet employs a RNN and its variant structures. Experimental results indicate that the GRU effectively mitigates the issue of information loss in long sequences while maintaining comparable performance to standard RNNs. Therefore, TSNet adopts the GRU structure to efficiently aggregate spectral features.Furthermore, subsequent experiments demonstrate that preprocessing the RNN inputs with mean value computation before feeding them into the model enhances its performance, surpassing the results obtained with raw input data. This suggests that averaging within the probability distribution space of tea samples contributes to effective feature extraction. Consequently, TSNet combines the GRU structure with mean processing, merging their outputs to fully exploit the strengths of both sequential modeling approaches, thereby improving the predictive performance of the model.

The structure of the TSNet model, as illustrated in Fig. 5, is specifically designed for moisture prediction in tea spectrum datasets. The model consists of two core components: feature extraction and sequential processing. The feature extraction module identifies key spectral bands, while the sequential processing module integrates spectral features from multiple sequences to enhance data representation. Finally, the processed data is passed through a Linear layer to compute the moisture prediction results. TSNet leverages the self-attention mechanism of Transformer to capture positional dependencies in spectral data, while incorporating the simplicity, ease of training, and overfitting resistance of RNN to ensure both accuracy and robustness in moisture prediction.

After the tea spectrum data is fed into the model, it first undergoes computation through the improved self-attention mechanism and feedforward network, followed by residual connection and normalization processing. Subsequently, the data is passed into a GRU structure and subjected to mean calculation, generating two feature vectors of size 1×256, which are then processed with another round of residual connection and normalization. Since GRU processes sequential data linearly and retains long-term dependencies, only the output at the final time step is retained as the final result of the GRU module. Finally, the outputs from GRU and mean calculation are processed through three Linear layers, yielding a 1×1 moisture prediction value, which serves as the final output of the model.

As shown in Fig. 5, TSNet keeps the spectral embedding dimension at $D = 256$ throughout the attention/GRU blocks. The improved self-attention module uses multi-head attention with $\mathrm{H} = 4$ heads (each head dimension $d_h = D/H$). The feed-forward network adopts a two-layer MLP with hidden width $\mathrm{D}_{ff} = 512$. The sequential module is a single-layer GRU with hidden size $\mathrm{D}_h = 256$ and we keep the last-step state. Add&Norm refers to residual connection followed by **LayerNorm**. For regularization, dropout with rate $\mathrm{p} = 0.10$ is applied after attention/FFN projections, and Adam uses weight decay 1e-4.

During the computation within the self-attention mechanism and the feedforward network, the initial input tensor is defined as

$$\mathrm{X}_{\text{init}} \in \mathbb{R}^{1\times 7\times 256}, \tag{9}$$

where the dimensions correspond to the batch size, sequence length, and feature dimension, respectively.

In the proposed improved self-attention mechanism, each spectral data sample is considered as an independent sequence. Accordingly, the input dimension is reformulated as

$$\mathrm{X}_{\text{mod}} \in \mathbb{R}^{(B_1\times B_2)\times L\times D}, \tag{10}$$

where B_1 and B_2 denote the two batch sizes, L represents the sequence length, and D is the data (feature) dimension.

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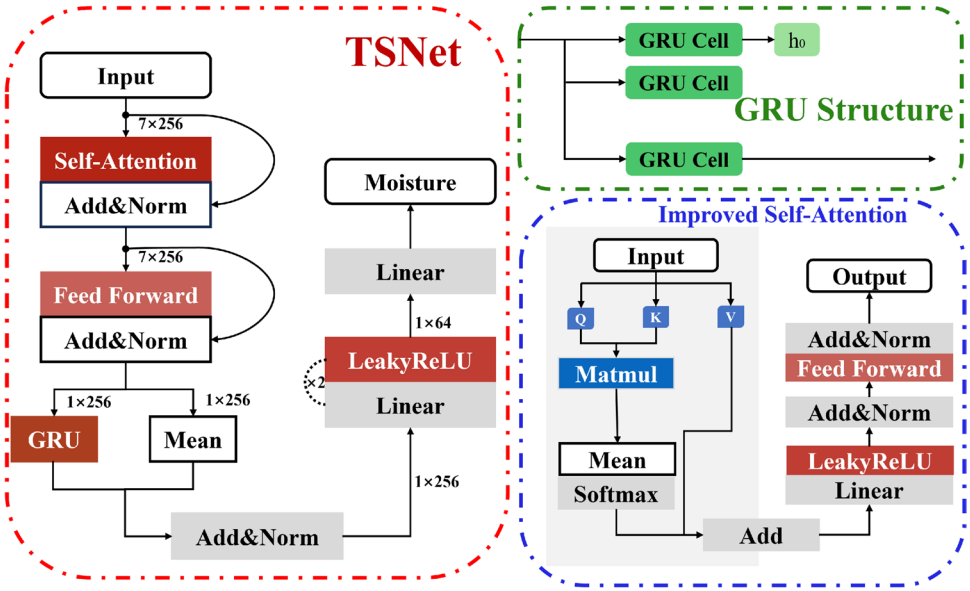


Fig. 5 TSNet Network Model Architecture

In practice, prior to inference, the input is reshaped using the view method as

$$X_{\text{reshaped}} \in \mathbb{R}^{1\times 7\times 256\times 1}. \tag{11}$$

Furthermore, all ReLU activation functions are replaced with LeakyReLU. Since ReLU zeros out negative inputs, it can lead to gradient vanishing, hinder network convergence, and result in uniform moisture predictions. In contrast, LeakyReLU maintains a small gradient for negative inputs, ensuring effective information propagation, which enhances both model stability and predictive performance.

Deployment design

Model serving

To meet an edge-first, cloud-backed deployment constraint, TSNet is packaged as a stateless inference microservice and exposed through a unified API (REST/gRPC). The model is exported in ONNX format so that edge nodes can execute FP16/INT8–quantized inference with ONNX Runtime for low tail-latency, while cloud replicas retain FP32 precision for fallback and batch analytics. Producer systems are decoupled from serving via an MQTT message bus, which simplifies fan-in from multiple probes and enables topic-based routing. The serving layer provides concurrency control (micro-batching with a default batch size of one to protect P99), health checking and circuit breaking (e.g., `/healthz` with timeouts and automatic failover), progressive rollout (canary/blue-green by device ID or topic), and end-to-end observability (P50/P95/P99 latency, throughput, error rate, and online MAE/RMSE). As shown in the middle panel of Fig. 6, TSNet Infer (ONNX Runtime) on the edge and TSNet Infer (Cloud) run in an active–standby arrangement: if an overheated edge node fails health probes, the router transparently diverts requests on the same MQTT topic to the cloud replica, preserving availability without sacrificing accuracy.

Real-time data pipeline

The end-to-end pipeline (Fig. 6) progresses layer by layer to maintain clear separation of concerns. In the Data Ingestion layer, spectra produced by the ATP8600 probe are captured by an edge collection agent, buffered and debounced, acknowledged, and enriched with timestamps plus device/batch identifiers before being published to MQTT. The stream then enters preprocessing, where dark/standard corrections and reflectance computation are performed according to Eq. 8, followed by band cropping, normalization, and quality control checks for saturation, SNR, and missing rate. Records that pass QC are tagged as qualified and written into a lightweight pull-based queue (a ring buffer with backpressure) for consumption by the serving tier.

In the Online Inference layer, qualified messages first reach a routing module that applies domain labels (e.g., tea type and integration time) to select the inference path. A black-tea batch measured at 10 ms, for example, is deterministically dispatched to the black-tea expert route, prioritizing TSNet Infer (ONNX Runtime) on the edge with seamless failover to TSNet Infer (Cloud) under overload or failed health checks. Predictions then pass through post-processing in the Results and Control layer: light smoothing and calibration are applied, predictive uncertainty is estimated, and process thresholds trigger events. High-uncertainty results

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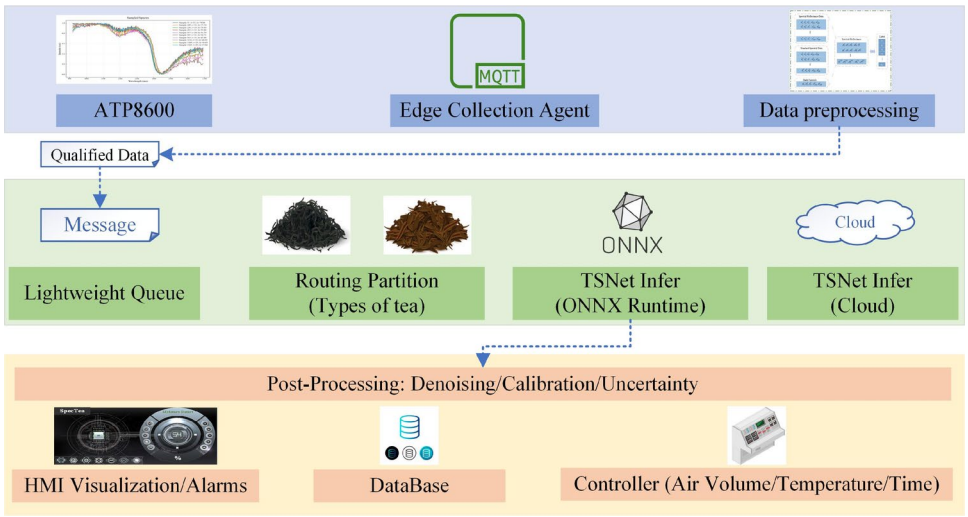


Fig. 6 Online deployment and dataflow of SpecTea. Top: Data Ingestion layer; Middle: Online Inference layer; Bottom: Results and Control layer

are flagged for re-inspection, raising HMI alerts rather than immediately driving actuators; confident excursions beyond thresholds initiate closed-loop control actions.

Partitioning

Partitioning is applied consistently across data, models, and compute to balance latency with cross-domain robustness. At the data level, tea type serves as the primary domain key and is augmented with device and integration-time metadata; per-domain statistics and online KPIs are maintained continuously. At the model level, tea-specific expert models are deployed with edge priority, while a general fallback model resides in the cloud; the router selects among these using domain labels, SLOs, and health signals, with optional online ensembling when beneficial. At the compute level, preprocessing and low-latency inference are fixed at the edge, whereas model selection, calibration, and retraining triggers run in the cloud. Outputs converge to three sinks in the Results and Control layer: HMI visualization/alarms, a persistent time-series database for audit and retraining reuse, and the industrial controller (air volume/temperature/time) for actuation—so that, for example, when post-processing confirms a moisture excursion with low uncertainty, the controller automatically adapts setpoints and all artifacts are persisted for retrospective evaluation and continual learning.

Due to the factory's confidentiality and safety policies, on-site photos/videos of the production line and control cabinets cannot be publicly disclosed. To ensure reproducibility and verifiability, we instead provide detailed deployment specifications together with time-stamped online operating logs used for the point-wise evaluation (N=4,611). We also clarify that the current deployment evidence is obtained from one factory site, and broader multi-factory validation is planned as future work.

Experiment

Experimental settings

All deep learning experiments were executed on a workstation with an Intel Core i7 CPU and an NVIDIA GeForce RTX 4090 GPU using the PyTorch framework with CUDA acceleration. To ensure reproducibility, we fixed the hardware/software stack across runs and maintained identical data pipelines and random seeds.Training employed the Adam optimizer with a learning rate of 2.5×10^{-4}, a batch size of 1, and 350 training iterations, using mean squared error (MSELoss) as the objective. Deep models were computed on the GPU, whereas traditional machine-learning baselines and preprocessing routines ran on the CPU. Given the dataset's limited numeric precision (two decimal places) and the modest computational complexity of tea-moisture prediction, any discrepancy arising from CPU/GPU floating-point arithmetic was considered negligible for the reported results.

All moisture labels used for model training/testing were obtained by a standardized drying-based reference method following the National Food Safety Standard GB 5009.3–2016 [20]. Moisture content was reported on a wet basis and computed as

$$\mathrm{MC}(\%) = \frac{m_{c+s} - m_{c+d}}{m_{c+s} - m_c} \times 100\%, \tag{12}$$

where m_c, m_{c+s}, and m_{c+d} denote the mass of the container, the container plus sample before drying, and the container plus dried sample, respectively. In our experiments, oven-drying was conducted using a halogen moisture analyzer (PC-16A, Shanghai Puchun, China) operated in automatic mode at $100^\circ\mathrm{C}$. Drying was terminated once

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the mass change was below 1 mg within 10 s (constant-mass criterion). According to the manufacturer, the measurement accuracy is $\pm 0.5\%$ MC. The key instruments and their specifications are summarized in Table 2.

Comparison of spectral preprocessing algorithm performance

To assess which preprocessing schemes are most suitable for tea–leaf NIR spectra, we benchmark eight common methods using PLSR for moisture prediction [21]. For each method, we compute the mean spectrum per sample group and standardize features with StandardScaler to improve comparability across bands. We then train PLSR models over a sweep of PCS and retain the best configuration per method. Performance is reported on calibration and test sets using the correlation coefficient (r), RMSE and RPD; detailed results appear in Table 3.

Table 3 shows that most preprocessing pipelines improve test-set RPD relative to the untreated baseline, whereas first- and second-derivative baselines underperform (RPD ≤ 1.8) and require few components (PCS < 10), indicating limited generalization. In our setting, numerical differentiation likely amplifies noise and converts absolute absorption structure into slope cues, making moisture-informative bands harder to recover and encouraging overfitting as PCS increases. By contrast, S–G smoothing, MSC scatter correction, and vector normalization achieve the strongest and most stable results: S–G yields the best test correlation among smoothers, MSC delivers the top performance within scatter-correction methods (highest r_p and lowest RMSEP), and vector normalization leads scaling approaches (test $r_p = 0.9884$, RMSEP $= 6.0000$, RPD $= 1.8826$). Based on these findings, we adopt a combined preprocessing routine(S–G smoothing/MSC/vector normalization) to enhance accuracy and robustness in subsequent modeling.

Table 2 Instruments and key specifications

Instrument	Model	Key specifications
NIR spectrometer	ATP8600	900–1700 nm; 256 channels.
Light source	ATG1100	360–2500 nm; 100 W halogen.
Fiber probe	MX2500-SIH600	Y-type; 6+1 core; 25.4° acceptance.
Moisture analyzer (ref.)	PC-16A	100°C auto-dry; accuracy ±0.5% (MC).

Comparative performance analysis of recurrent neural network architectures

We evaluate sequence-modeling choices for tea–leaf spectral data by embedding four architectures—vanilla RNN, LSTM, GRU, and Transformer—into an Elman-style framework to predict moisture content. Models are assessed on calibration and test sets using correlation (r), RMSE, and RPD, and we probe the effect of input formation with two preprocessing regimes: Normal (seven raw spectra per sample) and Mean (the per-sample average spectrum replicated to a 7×256 sequence). Results are summarized in Table 4.

Two consistent patterns emerge. First, averaging helps the simpler recurrent models: under Mean inputs, RPD improves relative to Normal by $+0.0889$ (RNN: 2.5335 vs. 2.4446), $+0.0913$ (LSTM: 2.5120 vs. 2.4207), and $+0.0274$ (GRU: 2.5173 vs. 2.4899). This suggests that averaging acts as a low-variance feature extractor that suppresses redundant or noisy variation the shallow recurrent dynamics would otherwise model poorly. Second, the Transformer shows the opposite but small effect: it performs marginally better on Normal than Mean inputs (RPD 2.5855 vs. 2.5707; $\Delta = +0.0148$), consistent with its stronger capacity to exploit fine-grained spectral cues that averaging may attenuate. In absolute terms, the Transformer attains the highest test-set RPD among all candidates (gains over RNN, LSTM, and GRU of $+0.1409$, $+0.1648$, and $+0.0956$, respectively), but this comes with a larger generalization gap and parameter footprint: the calibration–test RMSE gap is substantial under Normal

Table 3 Prediction results of moisture content using PLSR under different preprocessing methods

Category	Pretreatment Method	PCS	Calibration Set		Test Set		RPD
			r_c(%)	RMSEC(%)	r_p(%)	RMSEP(%)	
	Untreated	15	99.33	2.3878	0.9834	6.3475	1.8304
Smoothing	**S-G**	20	99.42	2.2057	0.9864↑	6.1359↓	1.8616↑
	WT	20	99.49	2.0803	0.9862	6.1514	1.8593
Scattering	**MSC**	12	99.66	1.7128	0.9897↑	5.8988↓	1.8987↑
	SNV	10	99.59	1.8738	0.9870	6.1262	1.8631
Baseline	1st Derivative	5	99.57	1.9091	0.9794	6.8707	1.7593
	2nd Derivative	8	99.77	1.3984	0.8487	13.0607	1.2760
Scaling	**Vector**	12	99.47	2.1281	0.9884↑	6.0000↓	1.8826↑
	Max-Min	15	99.80	1.2843	0.9885	6.1263	1.8631

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Table 4 Moisture prediction results of Elman neural network embedded with different sequential model structures

Data Types	Primary Model Architecture	Calibration Set		Test Set		RPD	Params
		r_c(%)	RMSEC(%)	r_p(%)	RMSEP(%)		
Normal	RNN	98.35	3.8249	0.9870	3.5586	2.4446	83KB
	LSTM	98.01	4.1223	0.9860	3.6292	2.4207	325KB
	GRU	98.21	3.9164	0.9874	3.4302	2.4899	244KB
	Transformer	99.54↑	2.1630↓	0.9898↑	3.1812↓	2.5855↑	1615KB
Mean	RNN	98.57	3.6471	0.9888	3.3132	2.5335	83KB
	LSTM	98.48	3.6152	0.9875	3.3702	2.5120	325KB
	GRU	98.52	3.6878	0.9885	3.3559	2.5173	244KB
	Transformer	99.35↑	2.9272↓	0.9916↑	3.2180↓	2.5707↑	1615KB

inputs (3.1812 vs. 2.1630; $\Delta = 1.0182$, reduced to 0.2908 with Mean), and the parameter count (1,615 KB) is roughly 5×–20× that of the recurrent baselines (83–325 KB). These observations indicate that while long-range attention improves peak accuracy, it is more prone to overfitting on short spectral sequences unless regularized or distilled. Conversely, LSTM's consistently lower RPD than RNN/GRU implies weak long-term dependencies in this task and suggests prioritizing mitigation of vanishing gradients (e.g., via GRU gating plus input averaging) over modeling very long temporal contexts. Overall, a GRU-based design augmented with mean pooling offers a favorable accuracy–efficiency–robustness trade-off, whereas Transformer variants warrant parameter reduction and stronger regularization if deployed under constrained conditions.

Performance analysis of TSNet for tea moisture prediction

We assess TSNet's accuracy on the tea–leaf spectral dataset against representative baselines from both traditional machine learning and recent deep models commonly used in tea–spectrum analysis. Because many comparators do not natively handle short spectral sequences, we standardize input dimensionality across methods: TSNet and the Elman-based sequence model consume a sequence tensor of size 7×256 (seven repeated spectra per sample), whereas non-sequential models receive a flattened vector of length 1×1792 obtained by concatenating the same seven spectra. Algorithms and their principal hyperparameters are listed in Table 5, ensuring a fair, reproducible comparison.

After extensive repeated runs, we selected the best configuration per model and recorded all outcomes. TSNet's predictions closely track ground-truth moisture values (Fig. 7(a)), and the training objective exhibits smooth, monotonic convergence (Fig. 7(b)), indicating stable optimization. Residual analyses on the calibration and test sets (Figs. 7(c)–(d)) show comparable dispersion and no systematic drift, suggesting that TSNet captures the salient spectral cues without overfitting to the calibration distribution. Taken together, these results support the effectiveness of combining attention-based band weighting with recurrent sequence aggregation for robust, sequence-aware moisture prediction on tea spectra.To further address reliability under limited sample size, we additionally performed a grouped 5-fold cross-validation at the batch level, where all seven repeated spectra from the same tea batch were always kept in the same fold to avoid information leakage. The cross-validated averaged performance of TSNet reaches $\mathrm{RMSE} = 2.7220\%$ and $\mathrm{MAE} = 2.2757\%$ (Table 6), which is consistent with the single-split evaluation and indicates that the reported accuracy is not dependent on a particular train/test partition.Regarding overfitting, we observe (i) smooth and stable optimization behavior during training (Fig. 7(b)), and (ii) comparable residual dispersion without systematic drift between calibration and held-out data (Figs. 7(c)–(d)). Together with the consistent 5-fold cross-validation results, these analyses suggest that TSNet does not exhibit evident overfitting on the current tea–spectrum dataset.

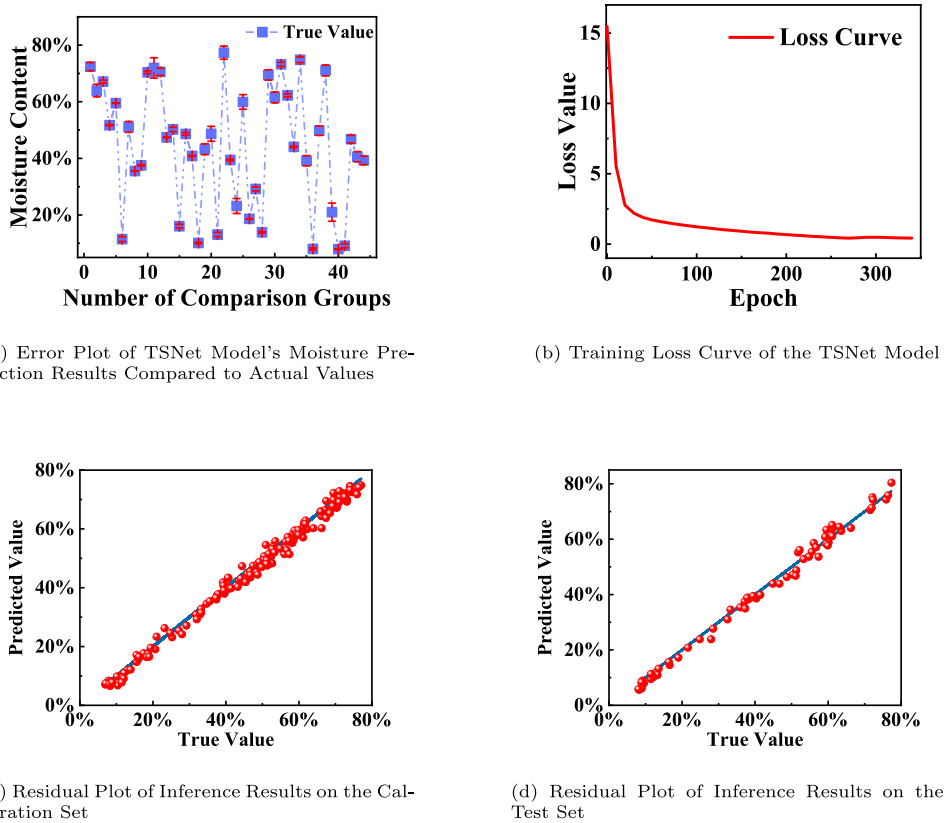
Table 5 Comparison of machine learning and deep learning models

Algorithm	Key Parameter Settings
PLSR	PCS = 15
LS-SVR	Kernel = Linear, C = 45
RFR	Trees = 7, Split Criterion = MSE
KNN	K = 14
BP-ANN [22]	Hidden_layers = 4
CNN [23]	Conv Layers = 2, Kernel Size = 8, Stride = 2
WOA-Elman [24]	Whales = 20, Max Iterations = 15

Comparison of TSNet with state-of-the-art models for moisture detection

To provide a comprehensive and objective evaluation, we compare TSNet against representative machine-learning and deep-learning baselines; detailed metrics are reported in Table 6. TSNet attains the best overall performance with $R = 0.9917$, $RMSE = 2.7612$, $MAE = 2.2776$,

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(a) Error Plot of TSNet Model's Moisture Prediction Results Compared to Actual Values

(b) Training Loss Curve of the TSNet Model

(c) Residual Plot of Inference Results on the Calibration Set

(d) Residual Plot of Inference Results on the Test Set

Fig. 7 Performance evaluation of the TSNet model on moisture prediction

$R^2 = 0.9831$, and $RPD = 2.7552$, exceeding the commonly used efficacy threshold of $RPD > 2.4$ and indicating strong quantitative agreement between predictions and ground truth. In contrast, classical machine-learning models (PLSR, LS-SVR, RFR, KNN) achieve $RPD < 2$, reflecting limited robustness under non-ideal acquisition conditions and higher spectral entropy. Among deep models, BP-ANN, CNN, and WOA-Elman improve upon machine learning, but remain behind TSNet on every metric; for example, CNN yields $RPD = 2.4433$ with a slightly lower relative error (RE $= 0.0727$) than TSNet, yet its higher $RMSE$ and lower R^2 point to weaker absolute accuracy and explained variance.

These results suggest that tasks characterized by spatial/temporal heterogeneity and moderate noise benefit from architectures with stronger representation capacity and sequence awareness. TSNet's attention-guided band weighting and recurrent aggregation jointly capture informative spectral structure, reducing absolute error while maintaining high correlation and explained variance. Consequently, within this evaluation setting, TSNet not only surpasses traditional machine-learning pipelines but also advances over recent deep models for tea-moisture detection, providing a more accurate and reliable predictor under realistic acquisition constraints.

Table 6 Comparative results of TSNet, traditional machine learning, and deep learning models on tea moisture detection

Cat.	Model	R(%)	RMSE(%)	MAE(%)	RE(%)	R^2(%)	RPD
Machine Learning	PLSR	98.83	6.0422	5.2285	0.2412	91.93	1.8761
	LS-SVR	98.64	6.1756	5.4839	0.2465	91.56	1.8556
	RFR	97.20	6.4784	5.3901	0.1678	90.72	1.8118
	KNN	97.56	5.9323	4.7580	0.1559	92.21	1.8934
Deep Learning	BP-ANN	98.46	3.4853	2.6492	0.0778	97.31	2.4701
	CNN	98.86	3.5621	2.7977	0.0727	97.19	2.4433
	WOA-Elman	98.55	3.7360	2.9929	0.0819	96.91	2.3858
	TSNet	99.17↑	2.7612↓	2.2776↓	0.0745	98.31↑	2.7552↑
	TSNet (Mean-5-fold)	99.20	2.722 ± 0.449	2.276 ± 0.389	0.0739	98.40	2.79

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Ablation study of TSNet components

To quantify the contribution of each design choice in TSNet, we conduct an ablation study by removing or replacing one component at a time while keeping the dataset split, training protocol, and hyperparameters unchanged. Table 7 summarizes the results. Overall, each component contributes positively to the final accuracy. In particular, removing the attention module (TSNet–Attn) increases RMSE from 2.7612 to 3.412 and MAE from 2.2776 to 2.801, indicating that band-wise dependency modeling is essential for suppressing redundant/noisy spectral responses. Removing the GRU-based sequence aggregator (TSNet–GRU) also degrades performance (RMSE 3.285, MAE 2.6912), confirming that sequence-aware fusion is important for leveraging the repeated spectra per sample. Furthermore, removing the mean branch (TSNet–Mean) yields a moderate drop (RMSE 2.9807, MAE 2.4529), suggesting that global aggregation provides complementary information to the recurrent path. Replacing LeakyReLU with ReLU (TSNet–ReLU) leads to inferior accuracy and stability, consistent with the risk of inactive neurons under low-variance spectral inputs. Finally, removing residual connections and normalization (TSNet–AddNorm) causes the largest degradation (RMSE 3.6503, MAE 3.0516), highlighting their critical role in optimization stability and generalization. These findings support that TSNet's final performance results from the synergy of attention-based band weighting, recurrent sequence modeling, and stabilized residual normalization.

Experimental analysis of the model for moisture detection in tea leaves at a tea factory

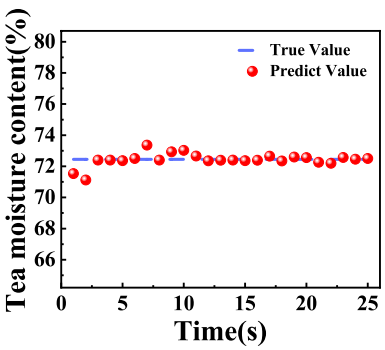
We conducted on-line validation in March 2024 at Hunan Hongda Tea Co., Ltd., using fresh Huangjincha leaves harvested to the "one bud and one leaf" standard from Dayutang Village, Chunhua Town, Changsha County (Hunan, China). The withering trough was instrumented with the proposed moisture-sensing platform, and predictions were streamed in real time. As illustrated in Fig. 8(a), a representative withering trace shows model estimates (red markers) closely tracking the batch-level reference steps (blue), with stable behavior around a typical operating point ($\approx 72.45\%$). Complementarily, Fig. 8(b) summarizes the distribution of measured/predicted moisture across batches (grouped by tea type), where the dashed red line indicates the target level (72.45%); the bulk of observations concentrate near this setpoint, supporting both pointwise fidelity and distributional consistency under factory conditions.

Deployable accuracy under industrial moisture ranges

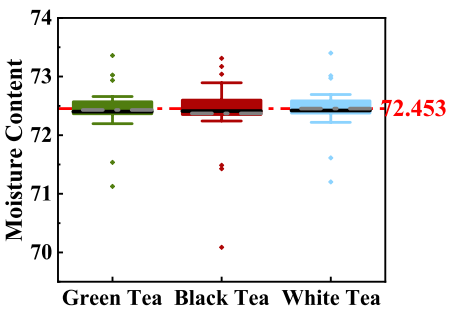
To mirror factory operating conditions, leaves were naturally withered to three target ranges, light ($70\% \le \mathrm{MC} < 80\%$), moderate ($60\% \le \mathrm{MC} < 70\%$), and heavy ($40\% \le \mathrm{MC} < 60\%$). Instead of reporting only group-level mean errors, we performed point-wise statistical evaluation over $N = 4611$ industrial

Table 7 Ablation study of TSNet. Each variant removes or replaces one component while keeping the same dataset split and training protocol

Cat.	Variant	R(%)	RMSE(%)	MAE(%)	RE(%)	R^2(%)	RPD
Ablation	TSNet–Attn	98.85	3.4120	2.8010	0.0785	97.10	2.4517
	TSNet–GRU	98.92	3.2850	2.6912	0.0774	97.25	2.5068
	TSNet–Mean	99.05	2.9807	2.4529	0.0758	97.80	2.6407
	TSNet–ReLU	98.98	3.1217	2.5619	0.0769	97.55	2.5806
	TSNet–AddNorm	98.70	3.6503	3.0516	0.0820	96.80	2.3517
	TSNet (full)	99.17↑	2.7612↓	2.2776↓	0.0745	98.31↑	2.7552↑



(a) Representative on-line prediction trace during withering (red: model predictions; blue: batch reference).



(b) Moisture distributions across batches/tea types; dashed red line denotes the target level (72.45%).

Fig. 8 Factory-floor performance of the proposed moisture-detection model

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samples (multiple time points and sampling locations across production batches). The overall point-wise accuracy reached $\mathrm{MAE} = 4.12\%$ with a bootstrap 95% confidence interval of $[3.05\%, 3.19\%]$ (2000 resamples), and $\mathrm{RMSE} = 3.96\%$. Residual analysis further indicates negligible systematic bias (mean residual $\mu = -0.0727$ for $\hat{y} - y$, equivalently $+0.071$ for $y - \hat{y}$), with residual standard deviation 3.96%.Band-wise statistics (Table 8) confirm stable performance across industrial moisture ranges, while the low moisture interval $[40\%, 60\%)$ exhibits a modestly higher error and wider dispersion, consistent with increased leaf structural heterogeneity in late withering. Fig. 9 summarizes the online predictions across tea types and moisture bands, and provides Bland Altman agreement analysis together with the absolute error boxplots for robust distributional inspection.These results indicate that the online system maintains stable, deployable accuracy across typical production moisture ranges.

Table 8 Point-wise error statistics across industrial moisture bands. Bootstrap 95% CIs are computed on MAE using 2000 resamples

Moisture band (%)	N(points)	MAE (%)	95% CI (%)	RMSE (%)	Bias (%)
[70,80)	1587	3.85	[3.73, 3.96]	3.66	-0.1702
[60,70)	1512	3.90	[3.78, 4.01]	3.60	-0.0428
[40,60)	1512	4.63	[4.48, 4.77]	4.55	-0.0002
Overall	4611	4.12	[4.05, 4.19]	3.96	-0.0727

Conclusion

We presented a non-destructive, real-time tea–moisture detection framework built on NIR spectroscopy and engineered for deployment on production lines. To counter spectral inconsistencies arising from leaf morphology and placement, we introduced a sigma adaptive sampling strategy that marries a five-point layout with Poisson disk separation and paired it with a dedicated acquisition setup to improve data robustness. On the modeling side, we developed a sequence architecture TSNet that combines self-attention for band weighting with GRU-based temporal aggregation to deliver accurate, stable moisture estimates from short spectral sequences. In controlled evaluations, TSNet surpassed both traditional machine-learning baselines and recent deep models, achieving a correlation coefficient of 0.9917, an RMSE of 2.7612, and an RPD of 2.7552 on the test set. For practical validation, we integrated TSNet into an online system and observed an average detection error of 3.89%, comfortably below the 5% threshold used in fresh tea quality control. Taken together, these results demonstrate industrial viability and indicate that the proposed pipeline can strengthen quality assurance and enable tighter process optimization. Future work will broaden validation across additional factories, tea cultivars, and processing stages, and will further enhance model generalization via domain adaptation under realistic variations in temperature, airflow, and operational conditions.

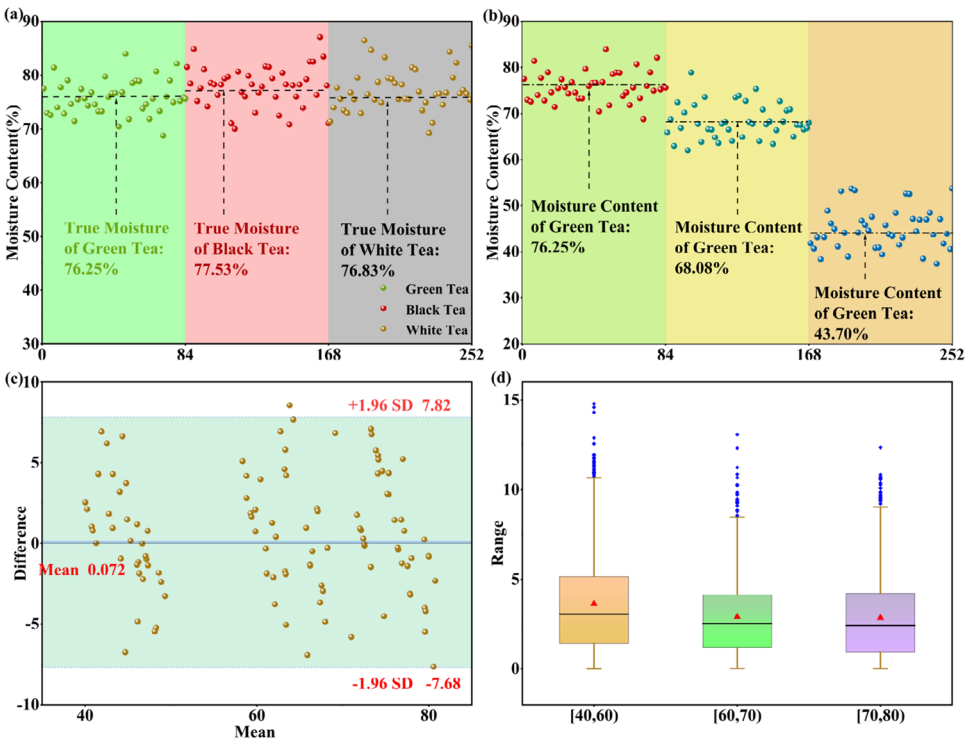


Fig. 9 Statistical evaluation of SpecTea for on-line tea-leaf moisture estimation under industrial conditions.(**a**) Point-wise on-line predictions for three tea types (green, black, and white tea) at the target moisture level; dashed lines indicate the corresponding batch-level reference moisture values.(**b**) On-line predictions for green tea under three industrial moisture bands (70–80%, 60–70%, and 40–60%); dashed lines denote the batch-level reference values for each band.(**c**) Bland–Altman analysis of predicted versus reference moisture values, where the central solid line denotes the mean bias and the dashed lines indicate the 95% limits of agreement (mean ± 1.96 SD).(**d**) Distribution of point-wise absolute errors across moisture bands shown as boxplots (box: interquartile range; center line: median; whiskers: 1.5×IQR; points: outliers)

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Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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